2+1 Dimensional Gravity and Chern-Simons Theory

X. D. H.

March 2, 2026

Abstract

This is a note on Chern-Simons formulation of 2+1 dimensional gravity, following the paper of Witten (1988)[1] .

Contents

1 Preliminaries I: Topology and Geometry

Here I will give a brief review of some basic concepts in topology that will be useful in understanding the Chern-Simons formulation of 2+1 dimensional gravity. It mainly follows the treatment in [2] However, it is not enough or rigorous for understanding any steps in Witten's paper, but serve as an introduction to many nouns he used in the paper.

1.1 Homeomorphisms and diffeomorphisms

Consider two topological manifold X and Y. We can define a **homeomorphism** between them:

> **Definition 1.** *A **homeomorphism** is a continuous bijection $f : X \to Y$ with a continuous inverse $f^{-1} : Y \to X$.*

If such a homeomorphism exists, we say that X and Y are **homeomorphic**, denoted as $X \cong Y$. We view the two manifold as "topologically equivalent", meaning they have the same topological properties.

If the manifolds are differential manifolds, we can further define a **diffeomorphism**:

> **Definition 2.** *A **diffeomorphism** is a smooth bijection $f : X \to Y$ with a smooth inverse $f^{-1} : Y \to X$.*

If we focus one 3 dimensional manifolds, there is an importants theorem:

> **Theorem 1.** *Moise's Theorem*
>
> *For 2 & 3 dimentional manifolds, being homeomorphic is equivalent to being diffeomorphic.*

This theorem implies that in 2 and 3 dimensions, the topological structure and the smooth structure are essentially the same. This is not true in higher dimensions, where there exist manifolds that are homeomorphic but not diffeomorphic (exotic spheres).

1.2 Mapping Class Group

1.2.1 Isotopy

We focus on the self-diffeo of one manifold $\mathcal{M}$, $f : \mathcal{M} \to \mathcal{M}$. We can define an **Isotopy** between two diffeomorphisms:

> **Definition 3.** *An **isotopy** between two diffeomorphisms $f, g : \mathcal{M} \to \mathcal{M}$ is a continuous family of diffeomorphisms $F(s) : \mathcal{M} \to \mathcal{M}$ for $s \in [0, 1]$ such that $F(0) = f$ and $F(1) = g$. If an Isotopy exists between f and g, we say that they are **isotopic**.*

Most diffeos considered in physics are isotopic to idenity map, yet there for topologically non-trivial manifolds, there can be diffeos that are not isotopic to identity.

- eg. **Dehn Twist** on torus T^2: Cut the torus along a non-contractible loop, twist one side by 360°, and then re-glue. This operation cannot be continuously deformed to the identity map.

1.2.2 Mapping Class Group

We can define an equivalence relation using Isotopy and define its equivalent class on the set of self-diffeos of $\mathcal{M}$:

> **Definition 4.** *The **Mapping Class Group** (MCG) of a manifold $\mathcal{M}$, denoted as $MCG(\mathcal{M})$, is the group of isotopy classes of self-diffeomorphisms of $\mathcal{M}$. .*
>
> *We can prove that the equivalent class set has a group sturcture.*

The mapping class group of 2 dimensional surfaces is well studies and they are generated by:

- **Dehn Twists** along non-contractible loops.

1.3 Fundamental Group

1.3.1 Definition

> **Definition 5.** *Homotopy Curve*
>
> *Consider a series of curves satisfying: $\gamma : [0,1] \to M, \quad \gamma(0) = x_0, \quad \gamma(1) = x_0$. If two curve γ_0, γ_1 can be continuously deformed to each other while keeping the endpoints fixed, we say they are **homotopic**, denoted as $\gamma_0 \simeq \gamma_1$.*
>
> *Mathematically, there exists a continuous map $F : [0,1] \times [0,1] \to M$ such that:*
>
> $$\begin{aligned} F(t,0) &= \gamma_1(t) \\ F(t,1) &= \gamma_2(t) \\ F(0,s) &= F(1,s) = x_0. \end{aligned} \tag{1}$$

> **Definition 6.** *Fundamental Group*
>
> *For a fixed base point x_0 in a topological space M, the **fundamental group** $\pi_1(M, x_0)$ is the set of equivalence classes of loops $[\gamma]$ based at x_0 under the homotopy relation.*

We can prove that the set of equivalence classes has a group structure, with the group operation defined as the concatenation of loops.

> **Theorem 2.** *For connected manifolds, the fundamental group is independent of the choice of base point up to isomorphism. Therefore, we often denote it simply as $\pi_1(M)$.*

1.3.2 More on Properties

> **Definition 7.** *Simply Connected*
>
> *A topological space M is said to be **simply connected** if it is path-connected and its fundamental group is trivial, i.e., $\pi_1(M) = \{e\}$, where e is the identity element.*

> **Theorem 3.** *Topological Invariance of Fundamental Group*
>
> *If two topological spaces M and N are homeomorphic, then their fundamental groups are isomorphic, i.e., $\pi_1(M) \cong \pi_1(N)$.*

Theorem 4. *Fully Characterization of 2D Surfaces by Fundamental Group*

The fundamental group fully characterizes the topology of closed 2-dimensional surfaces. Two closed surfaces are homeomorphic if and only if their fundamental groups are isomorphic.

1.4 Covering Space

1.4.1 Covering Space Definition

We can unwrap the non-contractible loops of a manifold by considering its covering space.

Definition 8. *Covering Space*

A ***covering space*** *of a topological space M is a topological space $\tilde{M}$ together with a continuous surjective map $p : \tilde{M} \to M$ such that*

- *For every point $x \in M$, there exists an open neighborhood U of x such that the preimage $p^{-1}(U)$ is homeomorphic to $U \times \Lambda$, where Λ is a discrete set.*

1.4.2 Topology of Covering Space

The surjective map used by defining the covering space can induce a homomorphism between the fundamental groups of the two spaces:

Theorem 5. *Induced Homomorphism on Fundamental Groups*

Let $p : \tilde{M} \to M$ be a covering map. Then, for any base point $\tilde{x}_0 \in \tilde{M}$ with $p(\tilde{x}_0) = x_0 \in M$, the map p induces a homomorphism $p_* : \pi_1(\tilde{M}, \tilde{x}_0) \to \pi_1(M, x_0)$.

This homoporphic is injective but not necessarily surjective. The image of p_* is a subgroup of $\pi_1(M, x_0)$ called the **characteristic subgroup** of the covering space. It only consist of paths that are lifted to $\tilde{M}$ to be loops instead of open curves.

Theorem 6. *For a connected manifold, any subgroup of its fundamental group can be realized as the characteristic subgroup of some covering space.*

Thus we can classify the covering spaces with the subgroups of the fundamental group.

Definition 9. *Universal Covering Space*

A ***universal covering space*** *of a topological space M is a covering space $\tilde{M}$ that is simply connected, i.e., its fundamental group is trivial: $\pi_1(\tilde{M}) = \{e\}$. which corresponds to the trivial subgroup of $\pi_1(M)$.*

1.5 Quotient Space

1.5.1 Quotient Space Definition

We can construct a new manifold by quotienting a manifold with a group action.

Definition 10. *Quotient Space*

Given a group G acting on a manifold M. The orbit of a point $x \in M$ under the action of G is the set $G \cdot x = \{g \cdot x | g \in G\}$. The ***quotient space*** *M/G is the set of all orbits of points in M under the action of G, equipped with the quotient topology.*

We can define a natural projection from the points in M to their corresponding orbits in M/G. which gives a quotient topology by defining open sets in M/G as those whose preimages under the projection are open in M.

- In general a Quotient Space **May NOT be a Manifold**.

Theorem 7. *Quotient Space as Manifold*
If a group G acts properly discontinuously and freely on a manifold M, then the quotient space M/G is also a manifold.

- **freely:** For any point $x \in M$, if $g \cdot x = x$ for some $g \in G$, then g must be the identity element of G. Equivanlently the isotropy group $G_x = \{g \in G | g \cdot x = x\}$ is trivial for all $x \in M$.
- **Propery Discontinuously:**
  - Any two points in different orbits have neighborhoods that do not intersect under the group action.
  - for any point x the isotropy group $G_x = \{g \in G | g \cdot x = x\}$ is finite.
  - For any point $x \in M$, there exists an open neighborhood U of x such that for all $g \in G$ not in the isotropy group G_x, the images $g \cdot U$ do not intersect with U, i.e., $g \cdot U \cap U = \varnothing$. for all $g \in G_x$, we have $g \cdot U = U$.

Definition 11. *Orbifold*
*If a group G acts properly discontinuously but not freely on a manifold M, the quotient space M/G is called an **orbifold**.*

An orbifold is a generalization of manifold with some mild singularities.

We can find a point on M to label the orbits in M/G, the complete set of such points is called a **fundamental region** of the group action.

1.5.2 Topology of Quotient Space

Theorem 8. *Topology of Quotient Space of Simply Connected Manifold*
If M is a simply connected manifold and G is a finite group then we can prove a relation between the fundamental group of the quotient space and the group action:

$$\pi_1(M/G) \cong G. \tag{2}$$

In this case the projection $p : M \to M/G$ is the universal covering map of the quotient space and M is the universal covering space of M/G.

Theorem 9. *Connected Manifold as Quotient Space*
Any connected manifold M can be represented as the quotient space of its universal covering space $\tilde{M}$ by the action of its fundamental group $\pi_1(M)$:

$$M \cong \tilde{M}/\pi_1(M). \tag{3}$$

Here we define the action of $\pi_1(M)$ on $\tilde{M}$ as: Mapping the loop in M mapped to open path in $\tilde{M}$ back by identifying its endpoints.

Mathematically, this action of $\pi_1(M)$ on $\tilde{M}$ is defined as a **Deck Transformation**:

Definition 12. *Deck Transformation*
For any $[\gamma] \in \pi_1(M)$ we can define a map $D_{[\gamma]} : \tilde{M} \to \tilde{M}$, that satisfy $p \circ D_{[\gamma]} = p$. It gives a representation of $\pi_1(M)$ as a group of homeomorphisms of $\tilde{M}$.

1.6 Geometization in 2D

Before we only consider topological manifolds, now we can further consider differential manifolds with riemannian metrics.

1.6.1 Standard Geometries in 2D

In 2 dimensions, there are three standard geometries with constant curvature:

- **Riemann Sphere** $\hat{\mathbb{C}} = \mathbb{C} \cup \infty$ with constant positive curvature metric:

$$ds^2 = \frac{4dzd\bar{z}}{(1+|z|^2)^2}. \tag{4}$$

- **Complex Plane** $\mathbb{C}$ with flat metric:

$$ds^2 = dzd\bar{z}. \tag{5}$$

- **Hyperbolic Plane** $\mathbb{H}^2 = \{z \in \mathbb{C} | \mathrm{Im}(z) > 0\}$ with constant negative curvature metric:

$$ds^2 = \frac{dzd\bar{z}}{(\mathrm{Im}(z))^2}. \tag{6}$$

1.6.2 Uniformization Theorem

We give out one of the most important theorems in 2D geometry:

Theorem 10. *Uniformization Theorem(Pure Topology Version)* {thm:Uniforr
Every Surface is homeomorphic (diffeomporphic) to D/Γ, where $D \in \{\hat{\mathbb{C}}, \mathbb{C}, \mathbb{H}^2\}$ and $\Gamma \subset Iso(D)$ is a properly discontinuously isometry on D.

This is a topological statement, we only say that a topological surface can be represented as a quotient space of one of the three standard geometries. However, with suitable operations we can give the quotient space a constant curvature metric induced from the standard geometry. [It is infact quite important to know this procedure though it seems to be doable]

Thus, we know that:

- Any closed surface can be given a constant curvature metric.
- Classification of such metric = Classification of isometry group actions on the standard geometries.

2 Preliminaries II: Vielbein Formalism of Gravity

{sec:Prelimi

Here I mainly follow the discussion in the appendix of [3].

2.1 Vielbein as non-coordinate Basis

2.1.1 Definition of Vielbein

In GR we know that the coordinate system on the manifold naturally generates a basis of the tangent space, which is called the coordinate basis. ∂_μ, dx^μ However, we can also choose other basis of the tangent space, we define a general othornomal basis as:

> **Definition 13.** *Othornormal Basis*
> *we define a set of basis vectors of the tangent space as e_a which satisfy:*
>
> $$g(e_a, e_b) = \eta_{ab} \tag{7}$$
>
> *where η_{ab} is depend on the spacetime signature. Equivantly, we can define a set of orthonormal basis of the cotangent space as θ^a which satisfy:*
>
> $$\theta^a(e_b) = \delta^a_b \tag{8}$$

Now we define the Vielbein as:

- Components of the coordinate basis in an orthonormal basis: $\partial_\mu = e_\mu{}^a e_a$
- Transformation matrix between the coordinate basis and the orthonormal basis

We can also define the inverse of the vielbein as: $e^\mu{}_a$ (note we always put the coordinate index first and the orthonormal index second), which satisfy:

$$e^\mu{}_a e_\mu{}^b = \delta^b_a, \quad e_\mu{}^a e^\nu{}_a = \delta^\nu_\mu. \tag{9}$$

We then can see that vielbein and inverse vielbein have the following property:

1. Can be used as basis transformation matrices between coordinate basis and orthonormal basis in tangent space and cotangent space.

$$e_a = e^\mu{}_a \partial_\mu, \quad \theta^a = e_\mu{}^a dx^\mu \quad \partial_\mu = e_\mu{}^a e_a, \quad dx^\mu = e^\mu{}_a \theta^a \tag{10}$$

2. The orthonormal condition can be written as:

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \quad \eta_{ab} = e^\mu{}_a e^\nu{}_b g_{\mu\nu} \tag{11}$$

Now we consider a general tensor field. A tensor can be written in the coordinate basis or the orthonormal basis or even mixed basis. For example, a vector field can be written as:

$$V = V^\mu \partial_\mu = V^a e_a \tag{12}$$

A tensor field can be written as:

$$T = T^{\mu\nu} \partial_\mu \otimes \partial_\nu = T^{ab} e_a \otimes e_b = T^{\mu a} \partial_\mu \otimes e_a \tag{13}$$

Moreover we can define a more general raising and lowering index rule:

- For a coordinate index, we can raise and lower it with the metric g.

- For an orthonormal index, we can raise and lower it with η.

This rule is exactly consistent with the definition of the vielbein, since we have:

$$e_\mu{}^a = g_{\mu\nu} e^\nu{}_b \eta^{ab}, \quad e^\mu{}_a = g^{\mu\nu} e_\nu{}^b \eta_{ab} \tag{14}$$

Finally we note that there are two other kinds of understanding of the vielbein:

- view $e_\mu{}^a$ as a set of one forms, which is just the basis vectors of the cotangent space θ^a so we may rename it as $e^a = e_\mu{}^a dx^\mu$.
- view $e^\mu{}_a$ as a (1,1) tensor field.

2.1.2 Basis Transformations GCT and LLT

Now we consider basis transformation, for a coordinate basis, the basis of the tangent space is tight to the coordinate system. Thus, it changes only when we choose a different coordinate system and the transformation matrix is given by the Jacobian of the coordinate transformation. We call this transformation as **GCT (General Coordinate Transformation)**.

However, for a orthonormal basis, we have the freedom of doing a basis transformation without changing the coordinate system. This is because:

$$e^a \to e'^a = \Lambda^a{}_b(x) e^b \tag{15}$$

is also a valid orthonormal basis as long as $\Lambda^a{}_b(x)$ is a local Lorentz transformation, i.e., it satisfies:

$$\Lambda^a{}_c(x) \Lambda^b{}_d(x) \eta_{ab} = \eta_{cd} \tag{16}$$

This basis transformation of the orthonormal basis is called **LLT (Local Lorentz Transformation)**. note that different from the lorentz transformation in the coordinate basis:

- LLT doesn't change the coordinate system, it only changes the basis of the tangent space.
- LLT is a local transformation, the transformation matrix can depend on the spacetime point instead of being a constant.

Generally, if we fix the orthonormal basis, then GCT will change the vielbein yet keep the orthonormal basis unchanged, while LLT is just a change of the orthonormal basis, thus it will change the vielbein but keep the coordinate basis unchanged.

2.2 Covariant Derivative

2.2.1 Spin Connection

Now we consider the covariant derivative of a vector field when we write it out in the orthormormal basis. To do this, we need to define a covariant derivative of the orthonormal basis vectors, which lead us to the spin connection:

Definition 14. *Spin Connection*

We define the spin connection $\omega_\mu{}^a{}_b$ as:

$$\nabla_\mu e_a = \omega_\mu{}^b{}_a e_b \quad \nabla_\mu e^a = -\omega_\mu{}^a{}_b e^b \tag{17}$$

In fact just as the normal connection, we can derive the covariant derivative of the cotangent space basis from that of the tangent space basis, we only need to define the first part.

With this definition, we can write the covariant derivative of a vector field component in the orthonormal basis as:

$$\nabla_\mu X^a{}_b = \partial_\mu X^a{}_b + \omega_\mu{}^a{}_c X^c{}_b - \omega_\mu{}^c{}_b X^a{}_c \tag{18}$$

We can prove a relation between the spin connection and the normal connection by evaluatingf the covariant derivative of a general vector field ∇X in both basis and compare the results. We have:

$$\Gamma^\nu_{\mu\lambda} = e^\mu{}_a \partial_\mu e_\lambda{}^a + e^\nu{}_a e_\lambda{}^b \omega_\mu{}^a{}_b \quad \omega_\mu{}^a{}_b = -e^\nu{}_b \partial_\mu e_\nu{}^a + e^\nu{}_b e_\lambda{}^a \Gamma^\lambda_{\mu\nu} \tag{19}$$ {relationcon

Then we evaluate the covariant derivative of Vielbein, if we view it as a 1-form, we have:

$$\nabla_\mu e_\mu{}^a = \partial_\mu e_\nu{}^a - \Gamma^\lambda_{\mu\nu} e_\lambda{}^a \tag{20}$$

We can plug in the relation between the spin connection and the normal connection, we have:

$$\nabla_\mu e_\nu{}^a = -\omega_\mu{}^a{}_b e_\nu{}^b \tag{21}$$

We can see that everything is constistent.

2.2.2 Spin Connection under LLT

Just as a normal connection spin connection is not a tensor. Yet due to the definition and linearity of the covariant derivative μ index, we can prove that:

- Under GCT the spin connection μ index transforms as a tensor. This means that we can view the spin connection as a one form in the coordinate basis with two indexes.
- Yet under LLT the spin connection doesn't transform as a tensor, we can prove from the definition or the property of the covariant derivative is a tensor under LLT that:

$$\omega_\mu{}^{a'}{}_{b'} = \Lambda^{a'}{}_a \Lambda^b{}_{b'} \omega_\mu{}^a{}_b + \Lambda^{a'}{}_c \partial_\mu \Lambda^c{}_{b'} \tag{22}$$

we shall notice that $\Lambda^{a'}{}_a$ and $\Lambda^b{}_{b'}$ are inverse transformation matrix of each other.

2.3 Cartan Structure Equations

Then comes the most important part of the vielbein formalism, which is the Cartan structure equations. These are equations that view geometric variables as differential forms and relates them to the vielbein and spin connection through exterior derivatives and wedge products.

We first define the vielbein one form and the spin connection one form as:

$$e^a = e_\mu{}^a dx^\mu, \quad \omega^a{}_b = \omega_\mu{}^a{}_b dx^\mu \tag{23}$$

Then we can rewrite the tortion tensor and the Riemann curvature tensor as differential forms as well:

$$T^a = \frac{1}{2} T^a{}_{\mu\nu} dx^\mu \wedge dx^\nu, \quad R^a{}_b = \frac{1}{2} R^a{}_{b\mu\nu} dx^\mu \wedge dx^\nu \tag{24}$$

where torsion tensor is defined as $T^a{}_{\mu\nu} = 2\Gamma^\lambda_{[\mu\nu]} e_\lambda{}^a$ and the Riemann curvature tensor is defined as $R^a{}_{b\mu\nu} = e^\lambda{}_b e_\rho{}^a R^\rho{}_{\lambda\mu\nu}$.

The Cartan structure equations are:

Theorem 11. *Cartan's structure equation*

$$T^a = de^a + \omega^a{}_b \wedge e^b, \quad R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b \tag{25}$$

The frist equation can be proved by directly using the relation between the spin connection and the normal connection eq. (19). The second is a bit more complicated.

If we take another exterior derivative of both sides of the Cartan structure equations, we can get the following identities:

$$dT^a + \omega^a{}_b \wedge T^b = R^a{}_b \wedge e^b, \quad dR^a{}_b + \omega^a{}_c \wedge R^c{}_b - R^a{}_c \wedge \omega^c{}_b = 0 \tag{26}$$

The first one is called the Bianchi identity for torsion, and the second one is just the normal Bianchi identity.

2.4 Christoffel Symbol

Above discussion we use general connection, now we consider the special case of the Christoffel symbol, which is the unique torsion free connection compatible with the metric.

We can see that:

- Torsion free condition means that:

$$de^a + \omega^a{}_b \wedge e^b = 0 \tag{27}$$

- Metric compatibility condition means that:

$$\nabla g = 0 \quad \Rightarrow \quad \nabla_\mu \eta_{ab} = 0 \tag{28}$$

  Thus we can get that:

$$\omega_{\mu ab} = -\omega_{\mu ba} \tag{29}$$

2.5 Extra: Convention and Definition of Differential Forms

Here list out some definitions for differential forms, in case I make a mistake in coefficients and signs. [I'm not sure what convention Witten uses. Yet its not conceptually important and I use proportional to instead of equal to in the main text]

3 Preliminary III: 2+1 Dimentional Classical Gravity

{sec:}

Here I'll talk about a standard set up in studying 2+1 dimensional gravity. We mainly follow the discussion in [2].

3.1 Local Structure

The Riemannian curvature of 2+1 dimensional gravity can be written directly in terms of Ricci Tensor and the metric:

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} + g_{\nu\sigma}R_{\mu\rho} - g_{\nu\rho}R_{\mu\sigma} - g_{\mu\sigma}R_{\nu\rho} - \frac{1}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})R \tag{30}$$

Thus if we consider the vaccum Einstein equation $R_{\mu\nu} = 0$, the Riemannian curvature vanishes identically! And if we consider the cosmological constant case $R_{\mu\nu} = \Lambda g_{\mu\nu}$, the Riemannian curvature becomes:

$$R_{\mu\nu\rho\sigma} = \Lambda(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}) \tag{31}$$

This is a Riemann Tensor of a constant convature spacetime.

- When $\Lambda = 0$, the spacetime is locally flat, i.e., locally isometric to Minkowski space $\mathbb{R}^{2,1}$.
- When $\Lambda > 0$, the spacetime is locally de Sitter space dS_3.
- When $\Lambda < 0$, the spacetime is locally anti-de Sitter space AdS_3.

3.2 Global Structure

However, being locally constant curvature spacetime doesn't mean that the theory is trivial, it may include various global properties. However, we shall ask what kinds of global topology may admit a constant curvature metric as well as a lorentzian structure(for physical spacetime is lorentzian).

The answer is given by the following theorem:

Theorem 12. *Let M be a compact 3 manifold with a flat, time-orientable Lorentzian metric and a purely spacelike boundary. Then M has the topology of:*

$$M \cong [0,1] \times \Sigma \tag{32}$$

Here Σ is a closed surface(compact and no boundary) homeomorphic to one of the boundary components of M.

Thus the topology of a physical spacetime is fixed by the topology of a initial spacelike slice Σ. In the following context we shall only focus on this kind of spacetime topology.

Remark:

In Witten's paper he uses $\mathbb{R}\times\Sigma$ topology, which is non-compact while $[0,1]\times\mathbb{R}$ is compact. Yet the difference needs further discussion. [But I don't understand it rigorously yet!!]

4 Chern-Simons Theory and Gravity (Witten)

{sec:chern-s

Here we continue on following Witten's discussion on the relation between 2+1 dimensional gravity and Chern-Simons theory [1].

4.1 Geometric Phase Space Analysis

4.1.1 Discussion Set Up

How to quantize a Theory, we usually start from analyzing the phase space of the theory and then construct a poisson braket of the phase space. Then one can transform the poisson bracket into commutator and get the quantum theory. Thus to study the quantum gravity in 2+1 dimensions, we need to first learn about the phase space of the classical 2+1 dimensional gravity.

The phase space of a classical theory is often defined as:

> **Definition 15.** *Phase Space*
> *The phase space is the space of all $q, \dot{q}$ at $t = 0$ surface.*

However, this definition is not covariant and we need tons of constraints to make it work, thus in 2+1 dimensional gravity, we use the following definition:

> **Definition 16.** *Phase Space*
> *The phase space is the space of all classical solutions to the equations of motion Modulo gauge transformations.*

For 2+1 dimensional gravity, this means that we want to find all the solutions to:

$$R_{\mu\nu} = 0 \tag{33}$$

To do this, we might first assume a spacetime topology of the manifold we want to consider, we focus on a manifold:

- With topology $M \cong \mathbb{R} \times \Sigma$. where Σ is a genus g closed surface.

We now simplify the problem of analyzing the phase space of 2+1 dimensional gravity to finding the space of all solutions to the vacuum Einstein equation $R_{\mu\nu} = 0$ on a manifold with topology $M \cong \mathbb{R} \times \Sigma$. These solutions can be found by quotienting the Minkowski space by a discrete subgroup of the isometry group of Minkowski space, which is the Poincare group $ISO(2, 1)$. In Witten's paper he gives out a analysis of those manifolds.

4.1.2 Manifolds with initial & final singularities

Witten explicitly constructs these class of solutions as examples, by quotienting the light cone of Minkowski, now we explain the procedure:

- **Preparation: Quotienting $\mathbb{H}$ to Σ_g**

  We can construct a 2-manifold with genus $g \geq 2$ by quotienting the hyperbolic plane.

  - **Step 1:** We first find the fundamental group of the surface Σ_g which is $\pi_1(\Sigma_g)$.
  - **Step 2:** We find a subgroup of $PSL(2, \mathbb{R})$ naming Γ which is isomorphic to $\pi_1(\Sigma_g)$ and acts **discretely** on $\mathbb{H}$.
  - **Step 3:** The quotient $\mathbb{H}/\Gamma$ defines a Riemann surface with genus g and a metric with constant negative curvature.

[In fact I don't understand how this step works mathematically, eg how to construct the metric and why we don't require the group to be free but only properly continuous (discrete), etc. I need to learn more about this step.]

> **Remark:**
> Note that here we quotient by a discrete subgroup other than a properly discontinuous subgroup, because the action of a properly discontinuous subgroup is equivalent to being discrete in the case of being a subgroup of $Aut(\mathbb{H}) = PSL(2,\mathbb{R})$.

Then we apply a similar procedure to the light cone of Minkowski space to construct a flat 3-manifold with topology $M \cong [0,1] \times \Sigma_g$ with initial singularity and final singularity.

To do this we first list out some useful facts about the 2+1 Minkowski spacetime:

- Normally we have the metric of:

$$ds^2 = -dt^2 + dx^2 + dy^2 \tag{34}$$

  And the light cone is defined by:

  - X^+ future light cone: $t^2 > x^2 + y^2, t > 0$
  - X^- past light cone: $t^2 > x^2 + y^2, t < 0$

- The isometry group is the 3 dimensional Poincare group $ISO(2,1)$ which is the semidirect product of the Lorentz group $SO(2,1)$ and the translation group $\mathbb{R}^{2,1}$.

  - The proper orthochronous Lorentz group $SO^+(2,1)$ is isomorphic to $PSL(2,\mathbb{R})$.

Quotienting the light cone to get a 3-manifold with initial & final singularities

We first notice that the light cone can be made up by hyperbolic planes:

$$t^2 = x^2 + y^2 + \tau^2 \tag{35}$$

where τ is a parameter for the hyperbolic plane.

- Each hyperbolic plane is isometric to $\mathbb{H}$ and the isometry group of the hyperbolic plane is the Lorentz group $SO^+(2,1)$ which is a subgroup of the Poincare group $ISO(2,1)$

Thus given a Σ_g we can construct a subgroup of the lorentz group Γ isometric to $\pi_1(\Sigma_g)$ and quotient the hyperbolic plane by this subgroup to get a Σ_g. If we do this quotient for the whole X^+ light cone we can get a 3 manifold with topology $M \cong [0,1] \times \Sigma_g$ and with an initial singularity at $\tau = 0$.

- The resulting manifold is a solution to the vaccum Einstein Field Equation for it is contructed by quotienting an isometry of Minkowski spacetime. This quotient group preserves the metric thus the resulting manifold is still flat.

  [I don't understand why?? It needs knowledge of metric under quotienting.]

- The resulting manifold has a $\Sigma_g \times [0,1]$ topology. This is because the lorentz group act the hyperbolic plane to itself due to its feature of preserving Minkowski metric. Thus, Γ as a subgroup of lorentz group act independently on each hyperbolic plane.

  Moreover, the isometry group of the hyperbolic plane is the lorentz group which is isomorphic to $PSL(2,\mathbb{R})$ and the hyperbolic plane is isometric to $\mathbb{H}$. Thus, quotienting the hyperbolic plane with the subgroup Γ of the lorentz group is equivalent to quotienting $\mathbb{H}$ with the same subgroup, which gives us a Σ_g. Thus, the resulting manifold has a $\Sigma_g \times [0,1]$ topology.

Parameters Counting

Finally, we can count how many distinct manifold can be constructed by this procedure, the answer depends on the choise of Γ subgroup. The final answer is that the moduli of genus g surface, which is $\mathbb{R}^{6g-6}$ gives us the label of distinct manifolds we can construct by this procedure.

[I don't understand this counting, it need some knowledge of Teichmuller space and moduli space of Riemann surface.]

4.1.3 General Flat Manifolds

Now we try to construct more general flat manifolds with topology $M \cong [0,1]\times\Sigma_g$ by quotienting more general subspace of the Minkowski space.

For a general flat 3-manifold $\hat{M}$, we can construct its simple connected universal cover M. Due to the flatness and being a universal covering space, M is isometric to the Minkowski space or a subspace of the Minkowski space. And by definition, $\hat{M}$ must be a quotient of M, which is a subspace of the Minkowski space, by a subgroup of the isometry group of the Minkowski space, which is the Poincare group $ISO(2,1)$.

Under this set up, we consider the fundamental group of $\hat{M}$, which is $\pi_1(\hat{M})$. Every element shall be a loop in $\hat{M}$, which can be lifted to a path in M. Then we can define a element of the Poincare group $ISO(2,1)$ $\phi(\gamma)$ which ensures that the path $[\gamma]$ in M which lifted from the loop $[\gamma]$ in $\hat{M}$, is mapped to a noncontractible loop after the quotienting by $\phi(\gamma)$.

Thus we can see that:

- Every flat 3-manifold give rise to a homomorphism from $\pi_1(\hat{M})$ to the Poincare group $ISO(2,1)$. It can be constructed by quotienting Minkowski space by the image of the homomorphism.

[but why this procedure can also cover quotienting a subspace of Minkowski space??? It only covers the case of quotienting the whole Minkowski space, but not the case of quotienting a subspace of Minkowski space, which is also a solution to the Einstein equation. I think perhaps this is one of the non-rigorous step in this derivation. Thus, we need a more rigorous field theoretical proof of the phase space.]

Thus, the phase space of 2+1 dimensional gravity can be identified with the following space:

$$\text{Phase Space} = \text{Hom}(\pi_1(\Sigma_g), ISO(2,1))/\sim \tag{36}$$

Here $\sim$ means that we quotient by the conjugation of the Poincare group $ISO(2,1)$. We know that if two homomorphisms ϕ_1,ϕ_2 are related as:

$$\phi_1(\gamma) = g\phi_2(\gamma)g^{-1} \quad \forall\gamma \in ISO(2,1) \tag{37}$$

Then they give rise to the same manifold under quotient.

If we want to consider the topology of $\hat{M} = \Sigma_g \times [0,1]$, then the fundamental group of the manifold is $\pi_1(\hat{M}) = \pi_1(\Sigma_g)$. Thus, the phase space of 2+1 dimensional gravity with topology $M \cong [0,1]\times\Sigma_g$ is:

{thm:Geophas

Theorem 13. *Geometric Analized Phase Space of 2+1 Dimensional Gravity*

The phase space of 2+1 dimensional gravity with topology $M \cong [0,1]\times\Sigma_g$ is:

$$\textit{Phase Space} = \textit{Hom}(\pi_1(\Sigma_g), ISO(2,1))/\sim \tag{38}$$

where $\sim$ means that we quotient by the conjugation of the Poincare group $ISO(2,1)$.

Parameter Counting

We then count the dimension of the phase space. For a surface Σ_g, its fundamental group can be generated by $2g$ generators a_i, b_i with one relation:

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \cdots, a_g, b_g | \prod_{i=1}^{g} a_i b_i a_i^{-1} b_i^{-1} = 1 \rangle \tag{39}$$

Thus a homomorphism from $\pi_1(\Sigma_g)$ to the Poincare group $ISO(2,1)$ is determined by the image of the generators a_i, b_i in the Poincare group $ISO(2,1)$. Which lead to $2g-1$ parameters with each parameter the dimension of the Poincare group $ISO(2,1)$, which is 6. Moreover, we need to quotient the phase space by the conjugation of the Poincare group $ISO(2,1)$, which lead to a reduction of 6 parameters. Thus, the dimension of the phase space is:

$$\dim(\text{Phase Space}) = (2g-1) \cdot 6 - 6 = 12g - 12 \tag{40}$$

> **Remark:**
>
> Finally, we shall note that this investigation is not complete and rigorous and as Witten have noted that we need a field theoretical proof of the phase space. Moreover, though we can construct the solutions, we also have left the problem that the quotienting may lead to manifold with undesirable properties, though being flat.
>
> In the final section section 5 we will view the phase space from a more rigorous CS field theory approach.

4.2 EH Action of Vielbein Formalism

The Vielbein formalism of gravity can be written from a action principle. We can prove that:

> **Theorem 14.** *Einstein-Hilbert Action in Vielbein Formalism*
>
> *The Einstein-Hilbert action in vielbein formalism in 4D can be written as:*
>
> $$S_{EH} = \frac{1}{2} \int_{\mathcal{M}} \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} \tag{41}$$

If we write this action in a normal integral (not differential form integral), we can get:

$$S_{EH} = \frac{1}{2} \int_{\mathcal{M}} d^4x \epsilon^{\mu\nu\rho\sigma} \epsilon_{abcd} {e_\mu}^a {e_\nu}^b {R_{\rho\sigma}}^{cd} \tag{42}$$

We can prove that viration of this action with respect to the vielbein ${e_\mu}^a$ and the spin connection ${\omega_\mu}^a{}_b$ gives us the Einstein equation and the torsion free condition respectively.

- Viration of ${e_\mu}^a$

We vary the action with respect to the vielbein ${e_\mu}^a$ and we can get:

$$\epsilon^{\mu\nu\rho\sigma} \epsilon_{abcd} {e_\nu}^b {R_{\rho\sigma}}^{cd} = 0 \tag{43}$$

Now we can exactly prove this is the Einstein field equation. Notice that:

$$\epsilon^{\mu\nu\rho\sigma} {e_\mu}^a {e_\nu}^b {e_\rho}^c {e_\sigma}^d = \epsilon^{abcd} \det(e) \tag{44}$$ {eq:epsilone

where $det(e)$ can be calculated by the following formula:

$$det(e)^2 det(\eta) = det(e^T \eta e) = det(g) \tag{45}$$

Thus we can get that:

$$det(e) = \sqrt{-det(g)} \tag{46}$$

By pulging eq. (44) into the viration equation and using the property of ϵ symbol, we can get:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \tag{47}$$

- Viration of ${\omega_\mu}^a{}_b$

The scheme is rather similar, we use the Cartan structure equation to rewrite the action in terms of ${\omega^a}_b$ explicitly and then vary the action with respect to ${\omega_\mu}^a{}_b$. The resulting equation is just the torsion free condition:

$$de^a + {\omega^a}_b \wedge e^b = 0 \tag{48}$$

or in component form:

$$D_\mu {e_\nu}^a - D_\nu {e_\mu}^a = 0 \tag{49}$$ {eq:torsionf

> **Remark:**
>
> Note that in fact we can not just assume this action to hold and say its equivalent to the Einstein gravity. Because in the deduction, we make an assumption that the vielbein is invertible, which is not generally guaranteed by the action itself (A general dynamical field are not assumed to have the component of a invertible matrix)
>
> However, in our following discussion we will not consider this problem. In modern topics, refer to [4], we may see that for AdS3 gravity, we can avoid the problem of ${e_i}^a$ being non-invertible by considering the Teichmuller Component of the phase space.

4.3 Chern-Simons Theory

A Chern-Simons Theory is given by a gauge field A associated to a gauge group G and on which we define a non-degenerate invariant bilinear form $Tr(*,*)$. The action of the Chern-Simons theory is given by:

$$S_{CS} = \int_{\mathcal{M}} \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right) \tag{50}$$

Here by $Tr A \wedge A \wedge A$ we mean $Tr(A_i, [A_j, A_k]) dx^i \wedge dx^j \wedge dx^k$ Thus, if we write the differential form integral into a normal integral, the action can be written as:

$$S_{CS} \propto \int_{\mathcal{M}} d^3x \epsilon^{\mu\nu\rho} \mathrm{Tr}\left(A_\mu(\partial_\nu A_\rho - \partial_\rho A_\mu) + \frac{2}{3} A_\mu [A_\nu, A_\rho]\right) \tag{51}$$

4.4 3D Gravity as Chern-Simons Theory

4.4.1 Motivation

Similar to the 4D case the EH action of 3D gravity in vielbein formalism can be written as:

$$S_{EH} = \int_{\mathcal{M}} \epsilon_{abc} e^a \wedge R^{bc} \tag{52}$$

if we write R^{ab} out using cartan structure equation, we see this resembles the Chern-Simons action. Thus we are motivated to rewrite the EH Action into a ISO(2,1) Chern-Simons action (To have the gauge group as the isometry group of Minkowski space, which is the Poincare group $ISO(2,1)$ is a natural choice). This rewriting consist of 2 step to go:

1. The EH Action can be written as a Chern-Simons Action with ISO(2,1) gauge group and suitable Killing form.
2. The gauge transformation of the Chern-Simons theory with ISO(2,1) gauge group is equivalent to the diffeomorphism and local lorentz transformation of vielbein and spin connection.

> **Remark:**
> In fact, the second step is not achivable for the whole diffeomorphism group, but only for small diffeomorphisms. For large diffeomorphisms, we need extra gauge transformations to cover.

4.4.2 Killing Form of ISO(2,1)

The first step we're facing is to find a non-degenerate invariant bilinear form of the Poincare group $ISO(2,1)$. For a general Poincare Group $ISO(d-1,1)$, in fact we don't have a non-degenerate invariant bilinear form, for we need a casimir like object to make it invariant and give out a number.

However, in the special case of $ISO(2,1)$, we can find one due to the property that the lorentz generator in 2+1 dimensions is J^{ab} a generator with two antisymmetric indices, in 3d its equivalent to a generator with one index J^a by using the epsilon symbol. Thus one can find a non-degenrate casimir:

$$W = \epsilon_{abc} P^a J^{bc} \tag{53}$$

And as a rersult we define: $J^a = 1/2\epsilon^{abc} J_{bc}$ and the Poincare algebra in 2+1 dimensions can be written as:

$$[J_a, J_b] = \epsilon_{abc} J^c, \quad [J_a, P_b] = \epsilon_{abc} P^c, \quad [P_a, P_b] = 0 \tag{54}$$

a non-degenerate invariant bilinear form:

$$\langle P_a, J_b \rangle = \delta_{ab} \quad \langle P_a, P_b \rangle = 0 \quad \langle J_a, J_b \rangle = 0 \tag{55}$$

> **Remark:**
> Here we use the notation that the P^a and J^{ab} generator can be raised and lower by the Minkowski metric η_{ab}. And so does the index of the ϵ_{abc}. But due to $det(\eta) = -1$, we have $\epsilon^{abc} = -\epsilon_{abc}$.
> [one should have check the exact definition of ϵ_{abc} and be careful if one want to perform detailed calculation. Yet its not conceptually important]

4.4.3 Rewriting EH Action into Chern-Simons Action

With these preparations, we can rewrite the EH action into a Chern-Simons action with gauge group $ISO(2,1)$ and the non-degenerate invariant bilinear form we just defined.

We first define a gauge field A as:

$$A_\mu = e_\mu{}^a P_a + \omega_\mu{}^a J_a \tag{56}$$

Here if we use the Christoffel symbol as the connection, the resulting spin connection $\omega_\mu{}^{ab}$ is antisymmetric in the indices a, b and thus we can write it as $\omega_\mu{}^a = 1/2\epsilon^{abc}\omega_{\mu bc}$ by using the epsilon symbol.

We note some useful facts about the gauge field A:

- **Gauge Transformation:** Consider a infinitesimal gauge transformation with the parameter:
$$u = \rho^a P_a + \tau^a J_a \tag{57}$$
is given by:
$$\delta_G A_\mu = -D^A_\mu u = -\partial_\mu u - [A_\mu, u] \tag{58}$$
Here δ_G stands for gauge transformation, later on we will see there are other transformations, eg. diffeomorphism and transformation generated by constriant in quantization. D^A_μ is the covariant derivative in the adjoint representation. If we write the above gauge transformation in terms of the vielbein ${e_\mu}^a$ and the spin connection ${\omega_\mu}^a$, we can get:
$$\delta_G {e_\mu}^a = -\partial_\mu \rho^a - \epsilon^{abc} e_{\mu b}\tau_c - \epsilon^{abc}\omega_{\mu b}\rho_c \tag{59}$$
{eq:gaugetra
$$\delta_G {\omega_\mu}^a = -\partial_\mu \tau^a - \epsilon^{abc}\omega_{\mu b}\tau_c \tag{60}$$
{eq:gaugetra

- **Curvature Tensor** The curvature tensor is given by:
$$\begin{aligned} F_{ij} &= [D_i, D_j] &(61)\\ &= P_a\Big(\partial_i e^a_j - \partial_j e^a_i + \epsilon^{abc}\big(\omega_{ib}e_{jc} + e_{ib}\omega_{jc}\big)\Big) &(62)\\ &+ J_a\Big(\partial_i {\omega_j}^a - \partial_j {\omega_i}^a + \epsilon^{abc}\omega_{ib}\omega_{jc}\Big). &(63)\end{aligned}$$

> **Remark:**
> In fact one can directly see that the two terms in the curvature tensor are just the EoM by viration of the EH action with respect to the spin connection and the vielbein respectively. Thus, the curvature tensor is zero, which is a standard Chern-Simons EoM, is equivalent to the EoM of the EH action.

With these preparations, we can prove the Chern-Simons action with the above gauge field is exactly the EH action of 3D gravity in vielbein formalism.
$$\begin{aligned} S &= \int_{\mathcal{M}} \mathrm{Tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right) &(64)\\ &= \int_{\mathcal{M}} \epsilon_{abc}e^a \wedge R^{bc} &(65)\\ &\propto \int_{\mathcal{M}} \epsilon^{ijk}\left(e_{ia}(\partial_j {\omega_k}^a - \partial_k {\omega_j}^a) + \epsilon_{abc}{e_i}^a{\omega_j}^b{\omega_k}^c\right) &(66)\end{aligned}$$

4.4.4 GCT and LLT as Gauge Transformation

Now we focus on the diffeomorphism of vielbein and spin connection.

- **Diffeomorphism:** Vielbein and spin connection can both be viewed as a 1-form thus the diffeomorphism is given by the Lie derivative. We consider a infinitesimal diffeomorphism generated by a vector fiel $-v^\mu$
$$\delta_D e^a = \mathcal{L}_{-v}e^a = -v^\mu(\partial_\mu {e_\nu}^a - \partial_\nu {e_\mu}^a)dx^\nu - \partial_\nu(v^\mu {e_\mu}^a)dx^\nu \tag{67}$$
$$\delta_D \omega^a = \mathcal{L}_{-v}\omega^a = -v^\mu(\partial_\mu {\omega_\nu}^a - \partial_\nu {\omega_\mu}^a)dx^\nu - \partial_\nu(v^\mu {\omega_\mu}^a)dx^\nu \tag{68}$$

- **Local Lorentz Transformation:** The local lorentz transformation generated by τ^a as a parameter is given by:
$$\delta_L e^a = \epsilon^{abc}e_b\tau_c \tag{69}$$
$$\delta_L \omega^a = d\tau^a + \epsilon^{abc}\omega_b\tau_c \tag{70}$$

Remark:

note that generally local lorentz transformation is generated by ω_{ab} antisymmetric parameter. However, in 3D we can rewrite it as $\omega_a = 1/2\epsilon_{abc}\omega^{bc}$ by using the epsilon symbol, thus we can write the local lorentz transformation in a more compact form.

These seems extremely distinct from the gauge transformation we defined before eq. (59). However, we can prove that they are equivalent.

τ^a transformation as LLT

First we notice that the gauge transformation generated by τ^a is exactly a LLT.

On shell ρ^a transformation as diffeomorphism

Yet the gauge transformation generated by ρ^a is not exactly a diffeomorphism. We explicitly write it out:

$$\delta_G e_\mu{}^a = -\partial_\mu \rho^a - \epsilon^{abc}\omega_{\mu b}\rho_c \tag{71}$$

$$\delta_G \omega_\mu{}^a = 0 \tag{72}$$

now we separately evaluate the difference between the gauge transformation generated by ρ and diffeomorphism generated by v^μ. We can see, if we take $\rho^a = v^\mu e_\mu{}^a$, the difference is given by:

$$\delta_G e_\mu{}^a - \delta_D e_\mu{}^a = -v^\nu(D_\nu e_\mu{}^a - D_\mu e_\nu{}^a) + \epsilon^{abc}v^\nu\omega_{\nu b}e_{\mu c} \tag{73}$$

We notice that the first term is just the torsion tensor, and on shell condition eq. (49) tells us that this is zero. And what we have left is a LLT generated by parameter: $\tau^a = v^\mu \omega_\mu{}^a$. Similarly we can prove that the diffeomorphism of the spin connection is also a LLT generated by the same parameter.

4.5 3D Grvaity as Chern-Simons Theory: Cosmological Constant Case

Now we want to consider the theory with cosmological constant added. An analogue with the above theory with zero cosmological constant, we can also rewrite the EH action with cosmological constant into a Chern-Simons action.

The EH action with cosmological constant can be written in the vielbein formalism as:

$$S_{EH} = \int_{\mathcal{M}} \epsilon_{abc}\left(e^a \wedge R^{bc} + \frac{\Lambda}{3} e^a \wedge e^b \wedge e^c\right) \tag{74}$$

$$\propto \int_{\mathcal{M}} \epsilon^{ijk}\left(e_{ia}(\partial_j \omega_k{}^a - \partial_k \omega_j{}^a) + \epsilon_{abc}e_i{}^a\omega_j{}^b\omega_k{}^c + \frac{\Lambda}{3}\epsilon_{abc}e_i{}^a e_j{}^b e_k{}^c\right) \tag{75}$$

Remark:

In fact this is not the standard way and Λ is not the usual cosmological constant. It is in fact $-\Lambda_{\text{canonical}}$. Here in this discussion we will follow witten's construction and note that $\Lambda < 0$ is dS and $\Lambda > 0$ is AdS, which is opposite to the standard convention. In the following discussion we will also follow this convention.

But now the gauge group instead of being the isometry group of Minkowski space, which is the Poincare group $ISO(2,1)$, is the isometry group of the spacetime with cosmological constant, which is $SO(2,2)$ for negative cosmological constant AdS and $SO(3,1)$ for positive cosmological constant dS.

The generators of the algebras can be written in a similar way:

$$[J_a, J_b] = \epsilon_{abc}J^c \quad [J_a, P_b] = \epsilon_{abc}P^c \quad [P_a, P_b] = \Lambda\epsilon_{abc}J^c \tag{76}$$

We can see that when the cosmological constant Λ is zero, the algebra reduces to the Poincare algebra in 2+1 dimensions. The killing form of the algebra can also be defined in a similar way:

$$\langle P_a, J_b\rangle = \delta_{ab} \quad \langle P_a, P_b\rangle = 0 \quad \langle J_a, J_b\rangle = 0 \tag{77}$$

4.5.1 Rewriting EH Action into Chern-Simons Action

By rewriting the gauge field A in a similar way:

$$A_\mu = e_\mu{}^a P_a + \omega_\mu{}^a J_a \tag{78}$$

> **Remark:**
> This rewriting is not taht inspiring in modern context, the next subsection I'll show that for AdS and dS we can rewrite the EH action into two Chern-Simons theory with gauge group $SL(2,\mathbb{R})$ and $SU(2)$ respectively, which is more useful in modern context.

We also note some useful facts of the gauge field A:

- **Gauge Transformation:** It resembles the original one but with an extra term in the transformation of $\omega_\mu{}^a$:

$$\delta_G e_\mu{}^a = -\partial_\mu \rho^a - \epsilon^{abc} e_{\mu b}\tau_c - \epsilon^{abc}\omega_{\mu b}\rho_c \tag{79}$$

$$\delta_G \omega_\mu{}^a = -\partial_\mu \tau^a - \epsilon^{abc}\omega_{\mu b}\tau_c - \Lambda\epsilon^{abc} e_{\mu b}\rho_c \tag{80}$$

- **Curvature Tensor** The curvature tensor is given by:

$$\begin{aligned} F_{ij} &= [D_i, D_j] && (81) \quad \texttt{\{eq:curvatur} \\ &= P_a\Big(\partial_i e_j^a - \partial_j e_i^a + \epsilon^{abc}\big(\omega_{ib}e_{jc} + e_{ib}\omega_{jc}\big)\Big) && (82) \quad \texttt{\{eq:curvatur} \\ &+ J_a\Big(\partial_i \omega_j{}^a - \partial_j \omega_i{}^a + \epsilon^{abc}\omega_{ib}\omega_{jc} + \Lambda\epsilon^{abc}e_{ib}e_{jc}\Big). && (83) \end{aligned}$$

Similarly, we can see that the curvature tensor is zero, which is a standard Chern-Simons EoM, is equivalent to the EoM of the EH action with cosmological constant. The torsion free condition is unchanged and the second EoM cahnge into the Einstein equation with cosmological constant.

We can also rewrite the EH action with cosmological constant into a Chern-Simons action with gauge group $SO(2,2)$ for $\Lambda > 0$ AdS and $SO(3,1)$ for $\Lambda < 0$ dS and the non-degenerate invariant bilinear form we just defined.

$$\begin{aligned} S_{CS} &= \int_{\mathcal{M}} \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A\wedge A \wedge A\right) && (84) \\ &\propto \int_{\mathcal{M}} \epsilon_{abc}\left(e^a \wedge R^{bc} + \frac{\Lambda}{3} e^a \wedge e^b \wedge e^c\right) && (85) \\ & && (86) \end{aligned}$$

4.6 3D Grvaity as Chern-Simons Theory: AdS Case

> **Remark:**
> In this subsection, we use the canonical cosmological constant convention Λ. Not the one used in above subsection.

In fact with the cosmological constant, we can redefine two gauge fields can rewrite the EH action into two Chern-Simons action with gauge group $SL(2,\mathbb{R})$ for AdS and $SU(2)$ for dS, which is more useful in modern context. To my corrent interest, I will focus on the AdS case. Discussion in this section mainly follows [5].

Here we turn back to the modern canonical way of writing the EH action, for AdS case $\Lambda = -1/l^2$, the EH action can be written as:

$$\begin{aligned} S_{EH} &= \frac{1}{16\pi G}\int_{\mathcal{M}} d^3x\sqrt{-g}(R + 2\Lambda) && (87) \\ &= \frac{1}{16\pi G}\int_{\mathcal{M}} \epsilon_{abc}\left(e^a \wedge R^{bc} + \frac{1}{3l^2} e^a \wedge e^b \wedge e^c\right) && (88) \end{aligned}$$

4.6.1 Rewriting EH Action into Two Chern-Simons Action

The gauge group $SO(2,2)$ is locally isomorphic to $SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$. Thus, we can rewrite the gauge field A into two gauge fields A^+ and A^- with gauge group $SL(2,\mathbb{R})$.

First the $SL(2,\mathbb{R})$ algebra can be written as 3 generators J_a with the following commutation relation:

$$[J_a, J_b] = \epsilon_{abc}\eta^{cd}J^d = \epsilon_{ab}{}^c J_c \tag{89}$$

and with a killing form:

$$\langle J_a, J_b \rangle = \eta_{ab} \tag{90}$$

Then we can define the two gauge fields A^+ and A^- as:

$$A^{+a} = \omega^a + \frac{1}{l}e^a \quad A^{-a} = \omega^a - \frac{1}{l}e^a \tag{91}$$

where the cosmological constant Λ is related to the AdS radius l by $\Lambda = -1/l^2$.

With these preparations, we can rewrite the EH action with cosmological constant into a Chern-Simons action with gauge group $SL(2,\mathbb{R})$ and the non-degenerate invariant bilinear form we just defined. We note:

$$S_{CS}[A] = \int_{\mathcal{M}} \mathrm{Tr}\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right) \tag{92}$$

Then we can prove (I'm not proving it here):

$$\begin{aligned} S_{CS} &= \frac{k}{4\pi}(S_{CS}[A^+] - S_{CS}[A^-]) && (93)\\ &= \frac{1}{16\pi G}\int_{\mathcal{M}} \epsilon_{abc}\left(e^a \wedge R^{bc} + \frac{1}{3l^2}e^a \wedge e^b \wedge e^c\right) && (94)\\ &= \frac{1}{16\pi G}\int_{\mathcal{M}} d^3x\sqrt{-g}(R + 2\Lambda) - \text{boundary term} && (95) \end{aligned}$$

Here to fit in the canonical normalization of the EH action, we have defined the level k as:

$$k = \frac{l}{4G} \tag{96}$$

With a standard Chern-Simons derivation, we can see that the EoM gives us the flatness condition of the gauge field A^+ and A^-:

$$F^+ = dA^+ + A^+ \wedge A^+ = 0 \quad F^- = dA^- + A^- \wedge A^- = 0 \tag{97}$$

4.6.2 GCT and LLT as Gauge Transformation

Here we still can prove the gauge transformation of the two Chern-Simons theory can give us a diffeomorphism and a local lorentz transformation.

I won't copy the proof here. The idea of [5] is to calculate the gauge transformation of $e_i{}^a$ by the gauge transformation of A^+ and A^- with parameter λ^+ and λ^- and given the relation between $e_i{}^a$ and $g_{\mu\nu}$ prove that its equivalent to a diffeomorphism of the metric:

$$\delta_G g_{\mu\nu} = \nabla_\mu \xi_\nu - \nabla_\nu \xi_\mu = \delta_D g_{\mu\nu} \tag{98}$$

5 Quantization and Phase Space Analysis (Witten)

{sec:Quantiz

Now we finally attempt to find the quantum theory. We only focus on the case of the mainfold have topology $M \cong \mathbb{R} \times \Sigma_g$ for some genus g surface Σ_g. We take the spactime coordinate x^0 to be the $\mathbb{R}$ direction and x^1, x^2 to be the spacial coordinates on Σ_g.

As we'll see the Lagrangian of the theory is a constraint system. Thus, we need to use some quantization procedure for constraint systems. Apart from the standard Dirac's procedure, here is more convenient to use another procedure, called the "constraints first quantization".

5.1 Constraint First Quantization

5.1.1 Classical Pre Constriant Phase Space

Having a Lagrangian we can try to quantize the theory. We follow the following steps:

1. **First order in ∂_0:** make sure all time derivative is in first order $\partial_0\phi$. If we have second order time derivative, we can introduce some auxiliary fields to make the Lagrangian first order. note that the Chern-Simons Lagrangian is already first order.
2. **Separate out Dynamical Variables:** separate variable with a time derivative and those without.
   - The ones with time derivative are the truly dynamical variables
   - The ones without time derivative are the Lagrange multipliers, whose EoM gives us some constraints.
3. **Find pre-constraint phase space and Poisson Bracket:** find out the canonical variables and the phase space. Define a Poisson Bracket on the phase space.

We then do it on the Lagrangian of 3D gravity with cosmological constant:

$$
\begin{aligned}
S = & -2\int \mathrm{d}t \int_{\Sigma} \epsilon^{ij}\, e_{i\alpha}\partial_0\omega_j^\alpha && (99)\\
& + \int \mathrm{d}t \int_{\Sigma} \Bigg[e_0{}^a\, \epsilon^{ij}\left(\partial_i\omega_j^a - \partial_j\omega_i^a + \epsilon^{abc}\omega_{ib}\omega_{jc} + \Lambda\, \epsilon^{abc} e_{ib}e_{jc}\right) && (100)\\
& \qquad + \omega_0{}^a\, \epsilon^{ij}\left(\partial_i e_j^a - \partial_j e_i^a + \epsilon^{abc}\left(\omega_{ib}e_{jc} + e_{ib}\omega_{jc}\right)\right)\Bigg]. && (101)
\end{aligned}
$$

Remark:
Here $i, j = 1, 2$ which are the spacial index. Thus we can view $e_i{}^a$ and $\omega_i{}^a$ as some 1-forms defined on the spacial slice Σ_g.

By looking at the Lagrangian, we can see many properties:

- **Constraints** $e_0{}^a$ and $\omega_0{}^a$ are not dynamical variables but lagrangian mulipliers that gives us the constraints.

  We can get the constriant equation by varying the action with respect to $e_0{}^a$ and $\omega_0{}^a$:

$$
\epsilon^{ij}\left(\partial_i\omega_j^a - \partial_j\omega_i^a + \epsilon^{abc}\omega_{ib}\omega_{jc} + \Lambda\, \epsilon^{abc} e_{ib}e_{jc}\right) = 0 \tag{102}
$$

$$
\epsilon^{ij}\left(\partial_i e_j^a - \partial_j e_i^a + \epsilon^{abc}\left(\omega_{ib}e_{jc} + e_{ib}\omega_{jc}\right)\right) = 0 \tag{103}
$$

  We can see that compare to eq. (81) that the constraint equation are the spacial part of the curvature tensor $F_{ij} = 0, i, j = 1, 2$. the constiants equation shows us that A_i on the spacial slice Σ_g is a flat connection.

- we can see that the $\omega_i{}^a$ and $e_i{}^a$ on the spacial slice Σ are the canonical variables and being the canonical conjugate of each other. Thus we can define a classical phase space and Poisson Bracket on it.

Theorem 15. *Pre-Constraint Phase Space and Poisson Bracket*

The Pre-Constraint Phase Space is space of configuration of $(e_i{}^a, \omega_i{}^a)$ on the spacial slice Σ_g.(resembling (p,q) configuration on a equal time slice in the standard Hamiltonian formalism)

The Poisson Bracket is defined as:

$$\{e_i{}^a(x), \omega_j{}^b(y)\} = \frac{1}{2}\epsilon_{ij}\eta^{ab}\delta^{(2)}(x-y) \tag{104}$$

$$\{e_i{}^a(x), e_j{}^b(y)\} = 0 \tag{105}$$

$$\{\omega_i{}^a(x), \omega_j{}^b(y)\} = 0 \tag{106}$$

This definition is consistent with the Lagrangian.

5.1.2 Constraint Quantization and Phase Space

In Dirac quantization, we use the pre-constraint phase space to quantize the theory, and view constriant as operator. The Hilbert space is the space of states that are annihilated by the constraint operator.

Here we would prefer to use another procedure, called the "constraint first quantization". The idea is that find a "True Phase Space" with the following steps and quantize the phase space directly:

1. **Find Solution Space of Constraint:** given may constraints $H_I, I = 1, 2, 3\dots$ find the solution space of the constraint equations $H_I = 0$. This is a subspace of the pre-constraint phase space.
2. **Find Equivalence Class:** the constraint H_I may generate a transformation on the canonical vairable (p,q) using the Poisson Bracket:

$$\delta_C q = \sum_I \epsilon_I\{H_I, q\} \quad \delta_C p = \sum_I \epsilon_I\{H_I, p\} \tag{107}$$

   We need to find the equivalence class of the solution space under this transformation, which is called the "True Phase Space". Or equivalently, the space of transformation invariant function on the solution space.
3. **Poisson Bracket on True Phase Space:** Poisson Bracket is unchanged, long as we only consider the transformation invariant function on the solution space, which is the "True Phase Space".

For our case of 2+1 dimensional gravity. The solution space of the constriant is the space of flat $G \in \{ISO(2,1), SO(3,1), SO(2,2)\}$ connection on the spacial slice Σ_g.

Remark:

we need to be careful to distinguish the phase space with the EoM solution space. The EoM solution space is the space of flat G connection on the whole spacetime. The phase space is the space of flat G connection on the spacial slice.

And amazingly we can find that the transformation generated by the constraint is just the G gauge transformation on the spacial slice.

Remark:
In fact in gauge theory, the constraint always generated the gauge transformation.

Thus we have the result:

Theorem 16. *True Phase Space of 2+1 Dimensional Gravity*
The True Phase Space of 2+1 dimensional gravity is the moduli space of flat $G \in \{ISO(2,1), SO(3,1), SO(2,2)\}$ *connection on the spacial slice* Σ_g*. Mathematically, this is equivalent to the following space:*

$$\mathcal{M} = \mathrm{Hom}(\pi_1(\Sigma_g), G)/\sim \tag{108}$$

where $\sim$ *is the equivalence relation given by the conjugation of* G *of group homomorphism* $\mathrm{Hom}(\pi_1(\Sigma_g), G)$*.*

{thm:True Ph

5.1.3 Compare with Geometric Analysis of Phase Space

We now construct a "True Phase Space" using the constraint first approach of the Chern-Simons formulation of 2+1 dimensional gravity theorem 16. We can compare it with the geometric analysis of the phase space we've done initially theorem 13. We can see that the result is **Exactly the Same!!!!**, apart form using different approaches.

However, we shall remark on something we have negelected which make 2+1 dimensional gravity not exactly equivalent to Chern-Simons Theory:

- Diffeomorphism is not exactly the gauge transformations. Gauge transformation only covers "small diffeomorphism", while "large diffeomorphism" is not included in the gauge transformation.
- In 2+1 Dimensional Gravity we assume that ${e_i}^a$ is invertible and want the metric to behave well. While the Chern-Simons theory doesn't care about this.

5.2 Real Polarization Quantization

Now we focus on quantizing the "True Phase Space". There are many approaches to quantize a phase space.

- If the phase space $\mathcal{M}$ is a cotangent bundle of some configuration space N then we can use a approach called **Real Polarization**

This is the true for Minkowski and dS case, but not for AdS case. This section we'll focus on the Minkowski case.

5.2.1 True Phase Space as Cotangent Bundle

A navie idea when looking at the pre-constraint phase space $({e_i}^a, {\omega_i}^a)$ is that we can view ${e_i}^a$ as the momentum and ${\omega_i}^a$ as the coordinate, thus the phase space is a cotangent bundle of the space of ${\omega_i}^a$ configuration.

But this is note generally true.

5.2.2 Hilbert Space

5.3 Kahler Polarization Quantization

For AdS case, the real polarization quantization can't work, thus we can follow another procedure that may work for both AdS, Minkowski and dS case, which is the Kahler polarization quantization. The idea is that:

- First we find a Kahler structure on the phase space.
- Then we find a holomorphic sector of a line bundle over the phase space and this may give us a Hilbert space.

This procedure is not discussed in Witten's paper, but it is widely used in the modern literature.

6 AdS3 Phase Space

{sec:AdS3 Ph

Here we follow the discussion in [4] to give an illustration of the modern perspective of the gravitational phase space of AdS3.

References

[1] E. Witten, “2 + 1 dimensional gravity as an exactly soluble system”, Nuclear Physics B **311**, 46–78 (1988).

[2] S. Carlip, *Quantum gravity in 2+1 dimensions*, Cambridge Monographs on Mathematical Physics (Cambridge University Press, Cambridge, 1998).

[3] S. M. Carroll, *Spacetime and geometry: an introduction to general relativity*, Pearson new international edition (Pearson Education, Harlow, 2014), 513 pp.

[4] L. Eberhardt, *Off-shell Partition Functions in 3d Gravity*, (Apr. 28, 2022) arXiv:`2204.09789 [hep-th]`, `http://arxiv.org/abs/2204.09789` (visited on 05/13/2025), pre-published.

[5] G. Compère and A. Fiorucci, *Advanced Lectures on General Relativity*, (Feb. 8, 2019) arXiv:`1801.07064 [gr-qc, physics:hep-th]`, `http://arxiv.org/abs/1801.07064` (visited on 03/13/2024), pre-published.